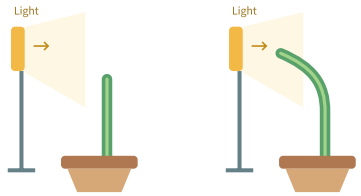
Light from the same side, three light-blocking treatments

The light is on the left for every setup. Before treatment, the coleoptiles（胚芽鞘） are similar in type, starting height and growth stage. Light intensity, exposure time, temperature and water supply are the same. The opaque（不透光） sleeve covers the whole part below the tip, leaving the tip exposed. The sleeve can bend as the plant grows and does not limit elongation（延長）.

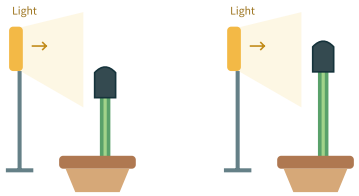
The animation speeds up growth and shows a typical response. The time and data are teaching simulations, not real measurements. The model does not show differences between individual plants.

A • No light blocked



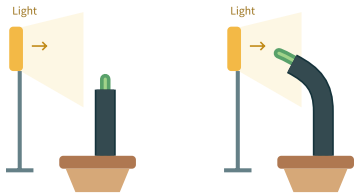
Before growth　After growth

B • Light blocked at the tip



Before growth　After growth

C • Light blocked below the tip



Before growth　After growth

My observation record

Setup	Treatment	First record • Bending direction	Revision (if needed)
A	No light blocked	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）
B	Light blocked at the tip	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）
C	Light blocked below the tip	（Bends left ／ Bends right ／ No clear bending）	（Bends left ／ Bends right ／ No clear bending）

Your observations are saved. If you repeat the experiment and revise them, your first observations and action record are also kept.

Use the main comparisons to draw conclusions

1. Compare A and B: after **blocking light at the tip**, what changes?（No clear bending towards the light. ／ It shows the same bending towards light.）

2. Compare A and C: when the lower part **does not** receive light directly, does the coleoptile（胚芽鞘） still bend?（It can still bend towards light. ／ No clear bending towards the light.）

3. Which part may **sense light from one side**?（Tip ／ Only the lower part）

04 • Extension 1

Does the tip also affect coleoptile（胚芽鞘） growth / elongation（延長）, as well as sensing light?

The light-blocking comparisons helped us confirm the part of the plant that senses light. Could the tip have another role? Add a new setup and change only whether the tip is present: compare intact A with D, whose tip has been removed.

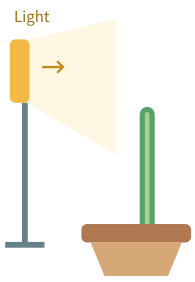
After removing the tip, what elongation（延長） do you predict compared with a complete coleoptile（胚芽鞘）?（Will not elongate（延長） ／ Similar elongation（延長） ／ More elongation（延長））

My reason

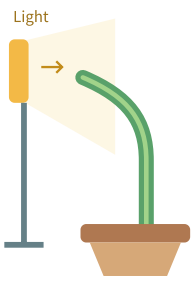
Your original extension 1 prediction and reason are saved before the first experiment. The animation and 24-hour period are teaching simulations.

A and D: compare before and after growth

A • No light blocked

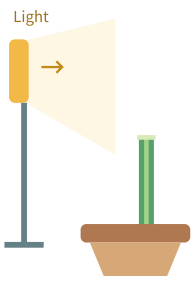


Before growth

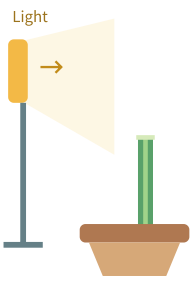


After growth

D • Tip removed



Before growth



After growth

Setup	Treatment	First record • Elongation（延長）	Revision (if needed)
A	No light blocked	（Yes ／ No）	（Yes ／ No）
D	Tip removed	（Yes ／ No）	（Yes ／ No）

Experimental conclusion

1. What does comparing elongation（延長） in A and D support?（The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）. ／ The presence of the tip does not affect the coleoptile（胚芽鞘）’s growth / elongation（延長）.）

The tip may produce and **pass down substance X**, promoting **elongation（延長）** below; this idea still needs to be tested.

04 • Extension 2

Can substance X transfer a growth-promoting effect? How does its position affect bending?

Researchers have prepared **agar（瓊脂） that has been in contact with coleoptile（胚芽鞘） tips** to test whether this effect can be **transferred** and how its position **affects bending**.

Extension 2 materials: oat coleoptiles（胚芽鞘） with tips removed × 4, growing containers × 4, blank agar（瓊脂） × 1, treated agar（瓊脂） × 3, timer × 1.

Both types of agar（瓊脂） look the same. **Blank agar（瓊脂） has not been in contact with** coleoptile（胚芽鞘） tips; **Treated agar（瓊脂） has been in contact with** coleoptile（胚芽鞘） tips. Some substances can enter and pass through agar（瓊脂）.

If **treated agar（瓊脂）** is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, how do you predict it will grow?（Bends left ／ Bends right ／ Stays fairly straight）

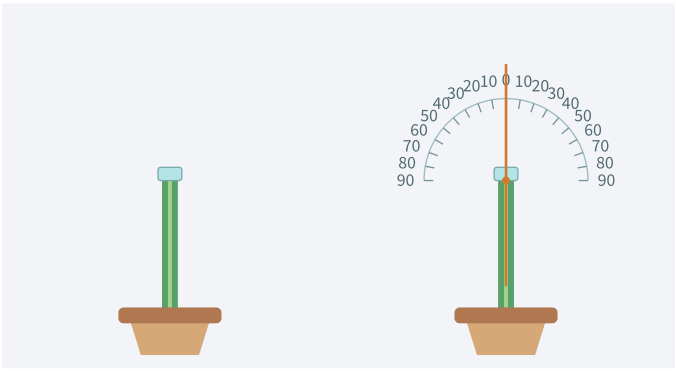
My reason

Four extension 2 setups in darkness

All coleoptiles（胚芽鞘） have their tips removed, start at a similar size and are kept in darkness. Agar（瓊脂） size, growing time, temperature and water supply are the same. You can view them before and after growth. Placing agar（瓊脂） on a coleoptile（胚芽鞘） does not physically hold it down.

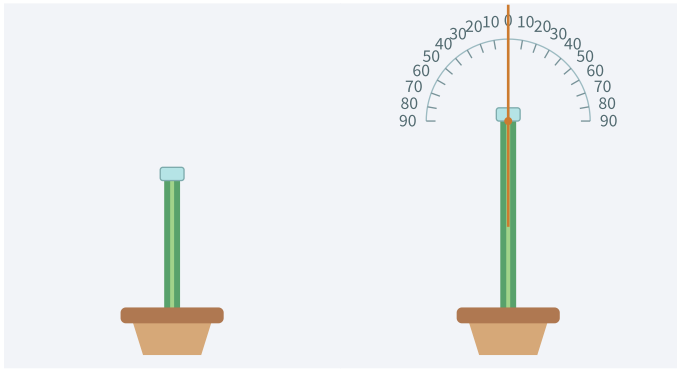
Growing time, gradual growth animations and angles are teaching simulations, not real measurements or exact experimental constants.

E • Blank agar（瓊脂） • Centre



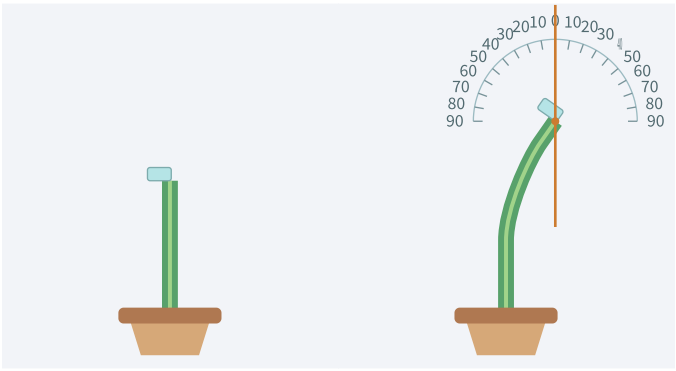
Darkness • Before growth　　Darkness • After growth

F • Treated agar（瓊脂） • Centre



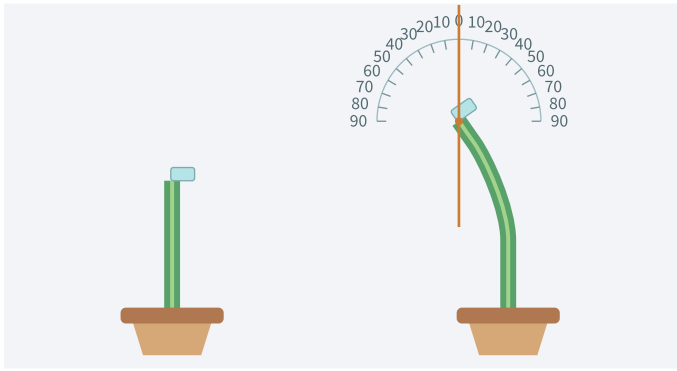
Darkness • Before growth　　Darkness • After growth

G • Treated agar（瓊脂） • Left



Darkness • Before growth　　Darkness • After growth

H • Treated agar（瓊脂） • Right



Darkness • Before growth　　Darkness • After growth

Setup	Treatment	Angle (°)	First record • Bending direction	Revision (if needed)
E	Blank agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）	_____° （Bends left ／ Bends right ／ No clear bending）
F	Treated agar（瓊脂） • Centre	_____°	（Bends left ／ Bends right ／ No clear bending）	_____° （Bends left ／ Bends right ／ No clear bending）
G	Treated agar（瓊脂） • Left	_____°	（Bends left ／ Bends right ／ No clear bending）	_____° （Bends left ／ Bends right ／ No clear bending）
H	Treated agar（瓊脂） • Right	_____°	（Bends left ／ Bends right ／ No clear bending）	_____° （Bends left ／ Bends right ／ No clear bending）

Analyse clues about substance X

1. What does the comparison of E and F support?（The tip may produce a growth-promoting effect that can be transferred. ／ All agar（瓊脂） produces the same growth-promoting effect.）

2. What does the comparison of F, G and H support?

◯ The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

◯ The position of treated agar（瓊脂） **does not** affect the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives （more ／ less） substance X, causing cells on the left to **elongate（延長）** （more ／ less） and the coleoptile（胚芽鞘） to bend to the （left ／ right） side.